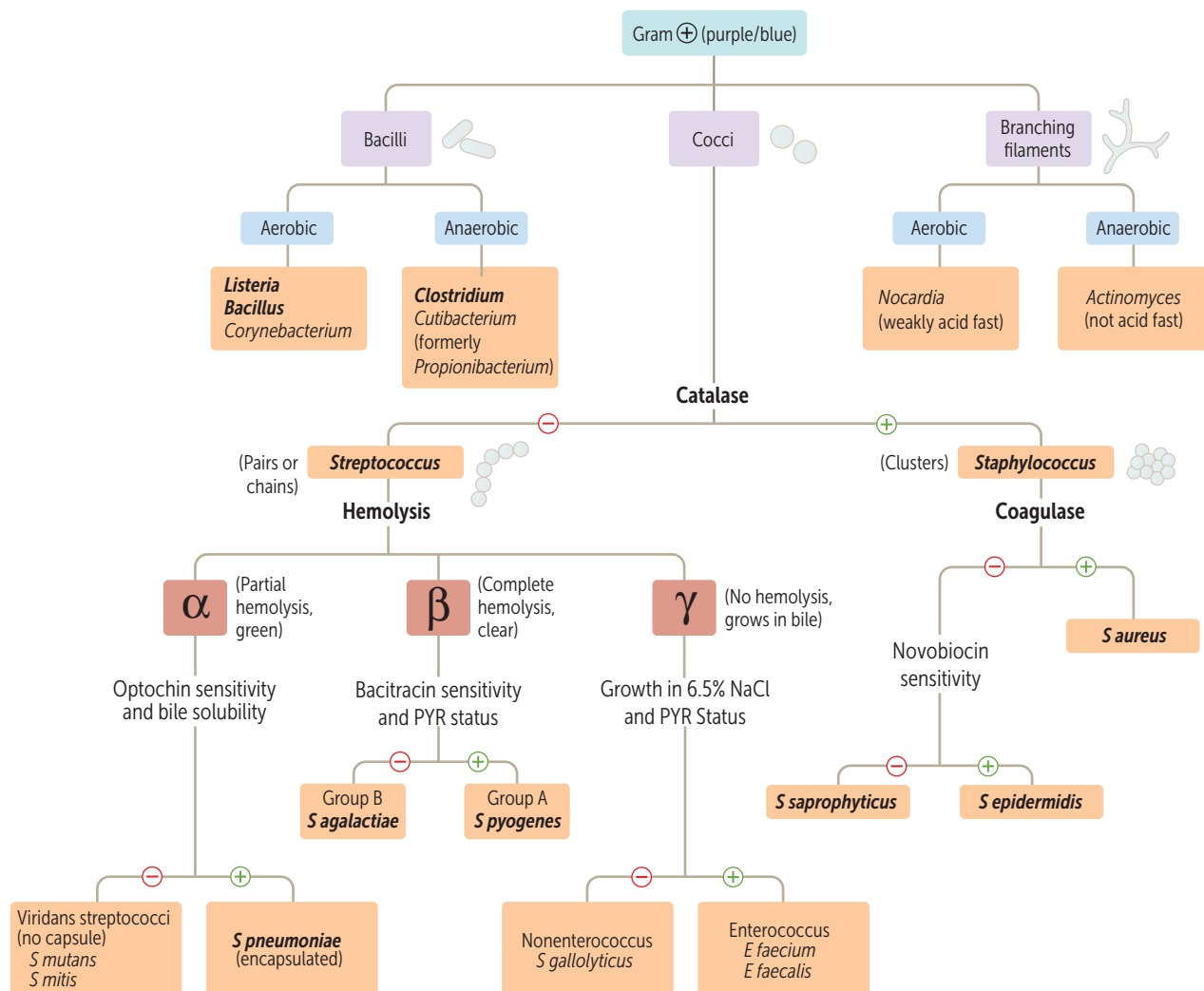


► MICROBIOLOGY—CLINICAL BACTERIOLOGY

Gram-positive lab algorithm



Important **tests** are in **bold**. Important **pathogens** are in **bold italics**.

Note: Enterococcus is either α- or γ-hemolytic.

PYR, Pyrrolidonyl aminopeptidase.

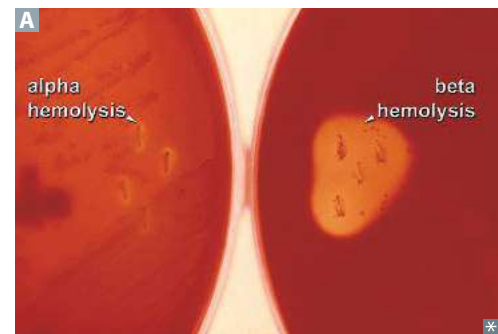
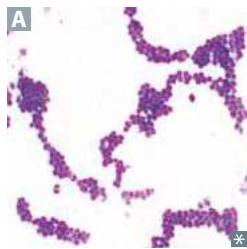
PK

Hemolytic bacteria **α -hemolytic bacteria**

Partial oxidation of hemoglobin → greenish or brownish color without clearing around growth on blood agar **A**.
Include *Streptococcus pneumoniae* and viridans streptococci.

 β -hemolytic bacteria

Complete lysis of RBCs → pale/clear area surrounding colony on blood agar **A**.
Include *Staphylococcus aureus*, *Streptococcus pyogenes* (group A strep), *Streptococcus agalactiae* (group B strep), *Listeria monocytogenes*.

***Staphylococcus aureus***

Gram $\oplus$, β -hemolytic, catalase $\oplus$, coagulase $\oplus$ cocci in clusters **A**. Protein A (virulence factor) binds Fc-IgG, inhibiting complement activation and phagocytosis. Commonly colonizes the skin, nares, ears, axilla, and groin.

Causes:

- Inflammatory disease—skin infections, organ abscesses, pneumonia (often after influenza virus infection), infective endocarditis, septic arthritis, and osteomyelitis.
- Toxin-mediated disease—toxic shock syndrome (TSST-1), scalded skin syndrome (exfoliative toxin), rapid-onset food poisoning (enterotoxins).

MRSA (methicillin-resistant *S aureus*)—important cause of serious healthcare-associated and community-acquired infections. Resistance due to altered penicillin-binding proteins (conferred by *mecA* gene). Some strains release Panton-Valentine leukocidin (PVL), which kills leukocytes and causes tissue necrosis.

TSST-1 is a superantigen that binds to MHC II and T-cell receptor, resulting in polyclonal T-cell activation and cytokine release.

Staphylococcal toxic shock syndrome (TSS)—fever, vomiting, diarrhea, rash, desquamation, shock, end-organ failure. TSS results in $\uparrow$ AST, $\uparrow$ ALT, $\uparrow$ bilirubin. Associated with prolonged use of vaginal tampons or nasal packing. Compare with *Streptococcus pyogenes* TSS (a toxic shock-like syndrome associated with painful skin infection).

S aureus food poisoning due to ingestion of preformed toxin → short incubation period (2–6 hr) followed by nonbloody diarrhea and emesis. Enterotoxin is heat stable → not destroyed by cooking.

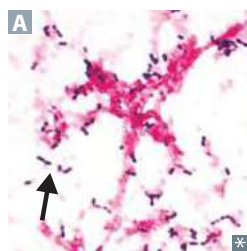
S aureus makes coagulase and toxins. Forms fibrin clot around itself → abscess.

Staphylococcus epidermidis

Gram $\oplus$, catalase $\oplus$, coagulase $\ominus$, urease $\oplus$ cocci in clusters. Novobiocin sensitive. Does not ferment mannitol (vs *S aureus*).
Normal microbiota of skin; contaminates blood cultures.
Infects prosthetic devices (eg, hip implant, heart valve) and IV catheters by producing adherent biofilms.

Staphylococcus saprophyticus

Gram $\oplus$, catalase $\oplus$, coagulase $\ominus$, urease $\oplus$ cocci in clusters. Novobiocin resistant.
Normal microbiota of female genital tract and perineum.
Second most common cause of uncomplicated UTI in young females (most common is *E coli*).

Streptococcus pneumoniae

Gram $\oplus$, α -hemolytic, lancet-shaped diplococci **A**.

Encapsulated. IgA protease. Optochin sensitive and bile soluble.

Most commonly causes **MOPS**:

- **M**eningitis
- **O**titis media (in children)
- **P**neumonia
- **S**inusitis

Pneumococcal pneumonia is associated with “rusty” sputum.

Patients with anatomic or functional hyposplenism or asplenia are predisposed to infection.

No virulence without capsule.

Pneumococcal vaccines are available in both conjugate (PCV13, PCV15, PCV20) and polysaccharide (PPSV23) formulations.

Viridans group streptococci

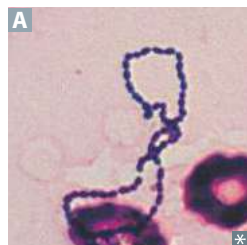
Gram $\oplus$, α -hemolytic cocci. Optochin resistant and bile insoluble. Normal microbiota of the oropharynx.

Streptococcus mutans and *S mitis* cause dental caries.

S sanguinis makes dextrans that bind to fibrin-platelet aggregates on damaged **heart** valves, causing infective endocarditis.

Viridans group strep live in the mouth, because they are not afraid **of-the-chin** (**op-to-chin** resistant).

Sanguinis = **blood**. Think, “there is lots of **blood** in the **heart**” (infective endocarditis).

***Streptococcus pyogenes* (group A streptococci)**

Gram $\oplus$ cocci in chains **A**. Group A strep cause:

- Pyogenic—pharyngitis, cellulitis, impetigo (“honey-crusted” lesions), erysipelas
- Toxicogenic—scarlet fever, toxic shock-like syndrome, necrotizing fasciitis
- Immunologic—rheumatic fever, glomerulonephritis

Bacitracin sensitive, β -hemolytic, pyrrolidonyl arylamidase (PYR) $\oplus$. Hyaluronic acid capsule and M protein inhibit phagocytosis. Antibodies to M protein enhance host defenses. Structurally similar to host proteins (ie, myosin); can lead to autoimmunity (ie, carditis seen in acute rheumatic fever).

Diagnose strep pharyngitis via throat swab, which can be tested with an antigen detection assay (rapid, in-office results) or cultured on blood agar (results in 48 hours).

“**Ph**”yogenes **ph**aryngitis can result in rheumatic “**p**hever” and glomerulone**ph**ritis.

Strains causing impetigo can induce glomerulonephritis.

Key virulence factors include DNase, erythrogenic exotoxin, streptokinase, streptolysin O. ASO titer or anti-DNase B antibodies indicate recent *S pyogenes* infection.

Scarlet fever—fine, blanching, generalized sandpaperlike rash sparing palms and soles, strawberry tongue, and circumoral pallor in the setting of group A streptococcal pharyngitis (erythrogenic toxin $\oplus$).

***Streptococcus agalactiae* (group B streptococci)**

Gram $\oplus$ cocci, bacitracin resistant, β -hemolytic, Group **B** for **B**abies!
 colonizes vagina; causes pneumonia, meningitis, and sepsis, mainly in **babies**.
 Polysaccharide capsule confers virulence.
 Produces CAMP factor, which enlarges the area of hemolysis formed by *S aureus*. (Note: CAMP stands for the authors of the test, not cyclic AMP.) Hippurate test $\oplus$. PYR $\ominus$.
 Screen pregnant patients at 35–37 weeks of gestation with rectal and vaginal swabs.
 Patients with $\oplus$ culture receive intrapartum penicillin/ampicillin prophylaxis.

Streptococcus gallolyticus

Formerly *S bovis*. Gram $\oplus$ cocci, colonizes the gut. Can cause bacteremia and infective endocarditis. Patients with *S gallolyticus* endocarditis have $\uparrow$ incidence of colon cancer.

Bovis in the **b**lood = **c**ancer in the **c**olon.

Enterococci

Gram $\oplus$ cocci. Enterococci (*E faecalis* and *E faecium*) are normal colonic microbiota that are intrinsically resistant to penicillin G and cause UTI, biliary tract infections, and infective endocarditis (following GI/GU procedures). Catalase $\ominus$, PYR $\oplus$, typically nonhemolytic.
 VRE (vancomycin-resistant enterococci) are an important cause of healthcare-associated infection.

Enterococci are more resilient than streptococci, can grow in 6.5% NaCl and bile (lab test).

Entero = intestine, *faecalis* = feces, *strepto* = twisted (chains), *coccus* = berry.

Bacillus anthracis

Gram $\oplus$, spore-forming rod that produces anthrax toxin, exotoxins consisting of protective antigen, lethal factor (inhibits MAP kinase $\rightarrow$ macrophage apoptosis), and edema factor (acts as adenyl cyclase $\rightarrow$ $\uparrow$ intracellular cAMP, upsetting homeostasis $\rightarrow$ edema, necrosis). Has a polypeptide capsule (poly D-glutamate). Colonies show a halo of projections, sometimes called “medusa head” appearance.

Cutaneous anthrax—painless papule surrounded by vesicles $\rightarrow$ ulcer with black eschar **A** (painless, necrotic) $\rightarrow$ uncommonly progresses to bacteremia and death.

Pulmonary anthrax—inhalation of spores, most commonly from contaminated animals or animal products, although also a potential bioweapon $\rightarrow$ flulike symptoms that rapidly progress to fever, pulmonary hemorrhage, mediastinitis (CXR may show widened mediastinum), and shock. Also called woolsorter’s disease. Prophylaxis with ciprofloxacin or doxycycline when exposed.

Both cutaneous and pulmonary anthrax may be complicated by hemorrhagic meningitis.

Bacillus cereus

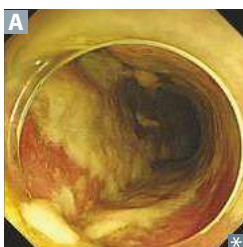
Gram $\oplus$ rod. Causes food poisoning. Spores survive cooking rice (reheated rice syndrome).

Keeping rice warm results in germination of spores and enterotoxin formation.

Emetic type causes nausea and vomiting within 1–5 hours. Caused by cereulide, a preformed toxin.

Diarrheal type causes watery, nonbloody diarrhea and GI pain within 8–18 hours.

Management: supportive care (antibiotics are ineffective against toxins).

Clostridioides difficile

Produces toxins A and B, which damage enterocytes. Both toxins lead to watery diarrhea → pseudomembranous colitis **A**. Often 2° to antibiotic use, especially clindamycin, ampicillin, cephalosporins, fluoroquinolones; associated with PPIs.

Fulminant infection: toxic megacolon, ileus, shock.

Difficile causes **diarrhea**.

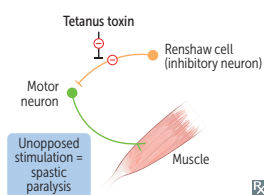
Diagnosed by PCR or antigen detection of one or both toxins in stool.

Treatment: oral vancomycin or fidaxomicin.

For recurrent cases, consider repeating prior regimen or fecal microbiota transplant.

Clostridia

Gram $\oplus$, spore-forming, obligate anaerobic rods. Tetanus toxin and botulinum toxin are proteases that cleave SNARE proteins involved in neurotransmission.

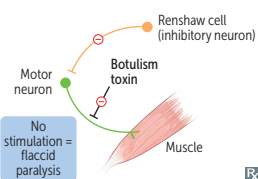
Clostridium tetani

Pathogen is noninvasive and remains localized to wound site. Produces tetanospasmin, an exotoxin causing tetanus. Tetanospasmin spreads by retrograde axonal transport to CNS and blocks release of GABA and glycine from Renshaw cells in spinal cord.

Causes **spastic** paralysis, trismus (lockjaw), risus sardonicus (raised eyebrows and open grin), opisthotonos (spasms of spinal extensors).

Tetanus is **tetanic** paralysis.

Prevent with tetanus vaccine. Treat with antitoxin +/- vaccine booster, antibiotics, diazepam (for muscle spasms), and wound debridement.

Clostridium botulinum

Produces a heat-labile toxin that damages SNARE proteins, thus preventing ACh release at the neuromuscular junction, causing botulism. In babies, ingestion of spores (eg, in honey) leads to disease (**floppy** baby syndrome). In adults, disease is caused by ingestion of preformed toxin (eg, in canned food).

Symptoms of botulism (the **5 D's**): **d**iplopia, **d**ysarthria, **d**ysphagia, **d**yspnea, **d**escending **f**laccid paralysis. Does not present with sensory deficits.

Botulinum is from bad **b**ottles of food, juice, and honey.

Treatment: human botulinum immunoglobulin. Local botulinum toxin A (Botox) injections used to treat focal dystonia, hyperhidrosis, muscle spasms, and cosmetic reduction of facial wrinkles.

Clostridium perfringens

Produces α -toxin (lecithinase, a phospholipase) that can cause myonecrosis (gas gangrene **A**; presents as soft tissue crepitus) and hemolysis. If heavily spore-contaminated food is cooked but left standing too long at $< 60^{\circ}\text{C}$, spore germinate → vegetative bacteria ingested → enterotoxin → late-onset (10–12 hours) food poisoning symptoms, resolution in 24 hours.

Perfringens **perforates** a gangrenous leg.

Spontaneous gas gangrene (via hematogenous seeding; associated with colonic malignancy) is most commonly caused by *Clostridium septicum*.

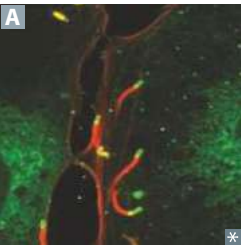
Corynebacterium diphtheriae



Gram $\oplus$ rods occurring in angular arrangements; transmitted via respiratory droplets. Causes diphtheria via exotoxin encoded by β -prophage. Potent exotoxin inhibits protein synthesis via ADP-ribosylation of EF-2, leading to possible necrosis in pharynx, cardiac, and CNS tissue. Symptoms include pseudomembranous pharyngitis (grayish-white membrane **A**) with lymphadenopathy (“bull’s neck” appearance). Toxin dissemination may cause myocarditis, arrhythmias, neuropathies. Lab diagnosis based on gram $\oplus$ rods with metachromatic (blue and red) granules and $\oplus$ Elek test for toxin. Toxoid vaccine prevents diphtheria.

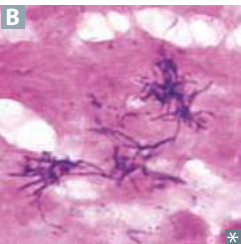
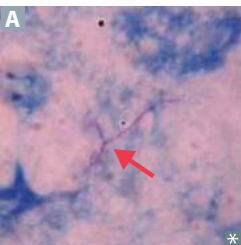
Coryne = club shaped (metachromatic granules on Löffler media).
Black colonies on cystine-tellurite agar.
ABCDEFGF:
ADP-ribosylation
 β -prophage
Corynebacterium
Diphtheriae
Elongation Factor 2
Granules
Treatment: diphtheria antitoxin +/- erythromycin or penicillin.

Listeria monocytogenes



Gram $\oplus$, facultative intracellular rod; acquired by ingestion of unpasteurized dairy products and cold deli meats, transplacental transmission, and by vaginal transmission during birth. Grows well at refrigeration temperatures (“cold enrichment”). Forms “rocket tails” (red in **A**) via actin polymerization that allow intracellular movement and cell-to-cell spread across cell membranes, thereby avoiding antibody. Listeriolysin generates pores in phagosomes, allowing its escape into cytoplasm. Characteristic tumbling motility in broth. Can cause amnionitis, sepsis, and spontaneous abortion in pregnant patients; granulomatosis infantiseptica; meningitis in immunocompromised patients, neonates, and older adults; mild, self-limited gastroenteritis in healthy individuals. Treatment: ampicillin.

Nocardia* vs *Actinomyces



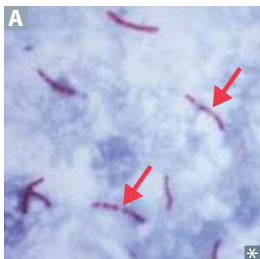
Both are gram $\oplus$ and form long, branching filaments resembling fungi.

Nocardia

Aerobe, catalase $\oplus$
Acid fast (weak) **A**
Found in soil
Causes pulmonary infections in immunocompromised (can mimic TB but with $\ominus$ PPD); cutaneous infections after trauma in immunocompetent; can spread to CNS $\rightarrow$ cerebral abscess
Treat with sulfonamides (TMP-SMX)
Treatment is a **SNAP**: Sulfonamides—*Nocardia*;

Actinomyces

Anaerobe, catalase $\ominus$
Not acid fast **B**
Normal oral, reproductive, and GI microbiota
Causes oral/facial abscesses that drain through sinus tracts; often associated with dental caries/extraction and other maxillofacial trauma; forms yellow “sulfur granules”; can also cause PID with IUDs
Treat with penicillin
Treatment is a **SNAP**: Sulfonamides—*Nocardia*; *Actinomyces*—Penicillin

Mycobacteria

Acid-fast rods (pink rods, arrows in **A**). Grows slowly in culture.

Mycobacterium tuberculosis (TB, often resistant to multiple drugs).

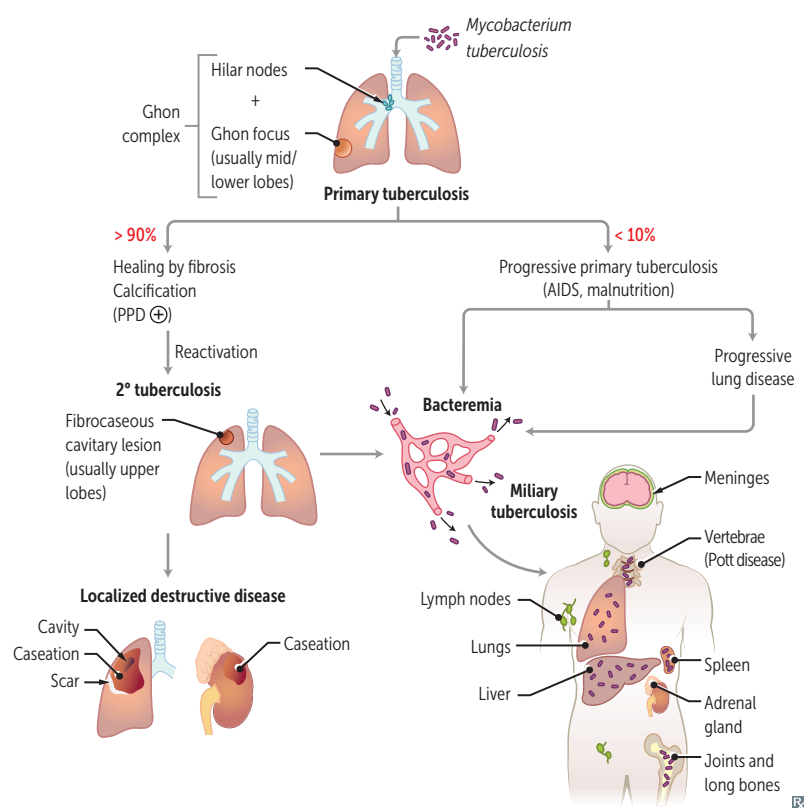
M avium–intracellulare (causes disseminated, non-TB disease in AIDS; often resistant to multiple drugs).

M scrofulaceum (cervical lymphadenitis in children).

M marinum (skin infection in aquarium handlers).

TB symptoms include fever, night sweats, weight loss, cough (nonproductive or productive), hemoptysis.

Cord factor creates a “serpentine cord” appearance in virulent *M tuberculosis* strains; activates macrophages (promoting granuloma formation) and induces release of TNF- α . Sulfatides (surface glycolipids) inhibit phagolysosomal fusion.

Tuberculosis

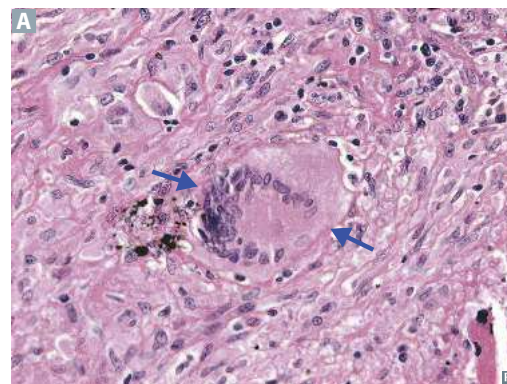
PPD $\oplus$ if current or past infection.

PPD $\ominus$ if no infection and in immunocompromised patients (especially with low CD4+ cell count).

Interferon- γ release assay (IGRA) has fewer false positives from BCG vaccination.

Caseating granulomas with central necrosis and Langhans giant cell (single example in **A**) are characteristic of 2° tuberculosis. Do not confuse Langhans giant cell (fused macrophages) with Langerhans cell (dermal APC).

TB reactivation risk highest in immunocompromised individuals (eg, HIV, organ transplant recipients, TNF- α inhibitor use). Reactivation has a predilection for the apices of the lung (due to the bacteria being highly aerobic).



Leprosy

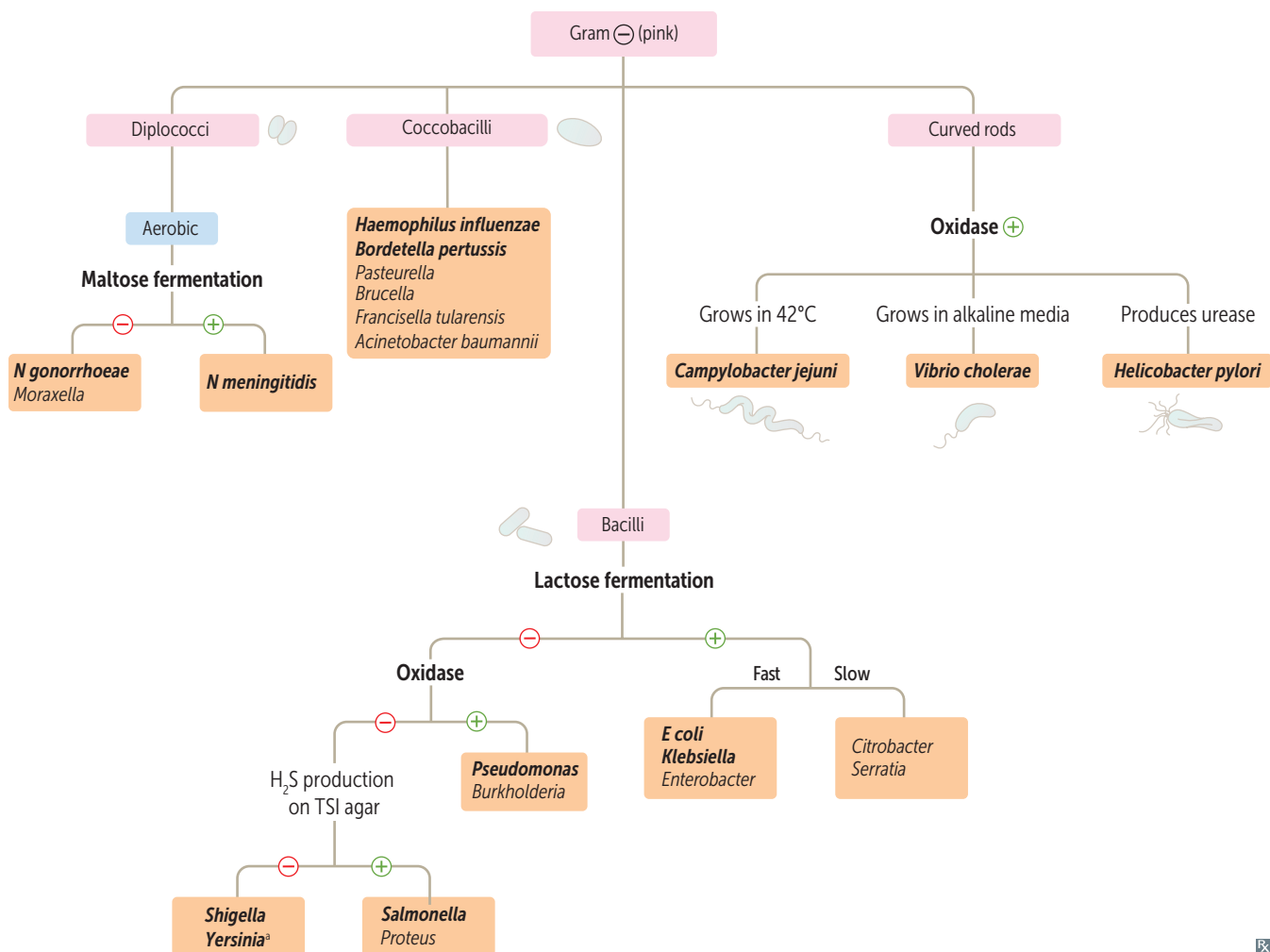
Also called Hansen disease. Caused by *Mycobacterium leprae*, an acid-fast bacillus that likes cool temperatures (infects skin and superficial nerves—"glove and stocking" loss of sensation) and cannot be grown in vitro. Diagnosed via skin biopsy or tissue PCR. Reservoir in United States: armadillos.

Leprosy has 2 forms (many cases fall temporarily between two extremes):

- **Lepromatous**—presents diffusely over the skin, with **leonine** (lionlike) facies **A**, and is communicable (high bacterial load); characterized by low cell-mediated immunity with a largely Th2 response. Lepromatous form can be **lethal**.
- **Tuberculoid**—limited to a few hypoesthetic, hairless skin plaques **B**; characterized by high cell-mediated immunity with a largely Th1 response and low bacterial load.

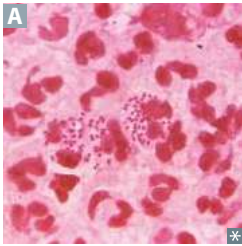
Treatment: **dapsone** and **rifampin** for tuberculoid form; **clofazimine** is added for lepromatous form.

Daring **rifles** target **cloaked leprechauns**.

Gram-negative lab algorithm

Important **tests** are in **bold**. Important **pathogens** are in **bold italics**.

^aPleomorphic rod/coccobacillus

Neisseria

Gram $\ominus$ diplococci. Metabolize glucose and produce IgA proteases. Contain lipooligosaccharides (LOS) with strong endotoxin activity.

N gonorrhoeae is often intracellular (within neutrophils) **A**.

Acid production: **m**eningococci—**m**altose and **g**lucose; **g**onococci—**g**lucose.

Gonococci

No polysaccharide capsule

No maltose acid detection

No vaccine due to antigenic variation of pilus proteins

Sexually or perinatally transmitted

Causes gonorrhea, septic arthritis, neonatal conjunctivitis (2–5 days after birth), pelvic inflammatory disease (PID), and Fitz-Hugh–Curtis syndrome

Diagnosed with NAAT

Condoms ↓ sexual transmission, erythromycin eye ointment prevents neonatal blindness

Treatment: single dose IM ceftriaxone; if chlamydial coinfection not excluded by molecular testing, add doxycycline

Meningococci

Polysaccharide capsule

Maltose acid detection

Vaccine (type B vaccine available for at-risk individuals)

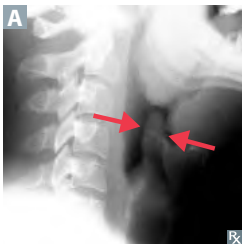
Transmitted via respiratory and oral secretions. More common among individuals in close quarters (eg, army barracks, college dorms)

Causes meningococcemia with petechial hemorrhages and gangrene of toes **B**, meningitis, Waterhouse-Friderichsen syndrome (acute hemorrhagic adrenal insufficiency)

Diagnosed via culture-based tests or PCR

Rifampin, ciprofloxacin, or ceftriaxone prophylaxis in close contacts

Treatment: ceftriaxone or penicillin G

Haemophilus influenzae

Small gram $\ominus$ (coccobacillary) rod. Transmitted through respiratory droplets. Nontypeable (unencapsulated) strains are the most common cause of mucosal infections (otitis media, conjunctivitis, bronchitis) as well as invasive infections since the vaccine for capsular type b was introduced. Produces IgA protease.

Culture on chocolate agar, which contains factors V (NAD⁺) and X (hematin) for growth; can also be grown with *S aureus*, which provides factor V via RBC hemolysis.

Haemophilus causes **e**piglottitis (endoscopic appearance can be “cherry red” in children; “thumb sign” on lateral neck x-ray **A**), **m**eningitis, **o**titis media, and **p**neumonia.

Vaccine contains type b capsular polysaccharide (polyribosylribitol phosphate) conjugated to diphtheria toxoid or other protein. Given between 2 and 18 months of age.

Does not cause the flu (influenza virus does).

Treatment: amoxicillin +/- clavulanate for mucosal infections; ceftriaxone for meningitis; rifampin prophylaxis for close contacts.

Burkholderia cepacia complex

Aerobic, catalase $\oplus$, gram $\ominus$ rod. Causes pneumonia in patients with underlying lung disease, such as cystic fibrosis. Often multidrug resistant. Infection is a relative contraindication to undergoing lung transplant due to its association with poor outcomes.

Bordetella pertussis

Gram $\ominus$, aerobic coccobacillus. Virulence factors include pertussis toxin (disables G_i), adenylate cyclase toxin ($\uparrow$ cAMP), and tracheal cytotoxin (destroys respiratory ciliated epithelium). Three clinical stages:

- Catarrhal—low-grade fevers, coryza.
- Paroxysmal—paroxysms of intense cough followed by inspiratory “whoop” (“whooping cough”), posttussive vomiting.
- Convalescent—gradual recovery of chronic cough.

Prevented by Tdap, DTaP vaccines (Tdap for preTeens and adulTs, DTaP for chilDren).

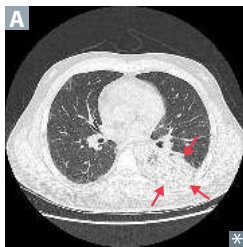
Produces lymphocytosis (unlike most acute bacterial infections).

Treatment: macrolides; if allergic use TMP-SMX.

Brucella

Gram $\ominus$, aerobic coccobacillus. Transmitted via ingestion of contaminated animal products (eg, unpasteurized milk). Survives in macrophages in the reticuloendothelial system. Can form non-caseating granulomas. Typically presents with undulant fever, night sweats, and arthralgia.

Treatment: doxycycline + rifampin or streptomycin.

Legionella pneumophila

Gram $\ominus$ rod. Gram stains poorly—use silver stain. Grow on charcoal yeast extract medium with iron and cysteine. Detected by presence of antigen in urine. Labs may show hyponatremia and elevated transaminases.

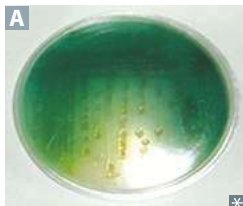
Aerosol transmission from environmental water source (eg, air conditioning systems, hot water tanks). Outbreaks associated with cruise ships, nursing homes. No person-to-person transmission.

Treatment: macrolide or quinolone.

Think of a French **legionnaire** (soldier) with his **silver** helmet, sitting around a campfire (**charcoal**) with his **iron** dagger—he is missing his **sister** (cysteine).

Legionnaires' disease—severe pneumonia (often unilateral and lobar **A**), fever, GI and CNS symptoms. Risk factors include older age, tobacco smoking, chronic lung disease.

Pontiac fever—mild flulike symptoms.

Pseudomonas aeruginosa

Aeruginosa—aerobic; motile, catalase $\oplus$, gram $\ominus$ rod. Non-lactose fermenting. Oxidase $\oplus$.

Frequently found in water. Increased virulence in acidic environments. Has a grapelike odor.

PSEUDOMONAS is associated with:

Pneumonia, **S**epsis, **E**cthyma gangrenosum, **U**TI, **D**iabetes, **O**steomyelitis, **M**ucoid polysaccharide capsule, **O**titis externa (swimmer's ear), **N**osocomial (healthcare-associated) infections (eg, catheters, equipment), **A**ddiction (injection drug use), **S**kin infections (eg, hot tub folliculitis, wound infection in burn victims).

Mucoid polysaccharide capsule may contribute to chronic pneumonia in patients with cystic fibrosis due to biofilm formation.

Produces **PEEP**: **P**hospholipase C (degrades cell membranes); **E**ndotoxin (fever, shock); **E**xotoxin A (inactivates EF-2); green **P**igment **A**.

Corneal ulcers/keratitis in contact lens wearers/minor eye trauma.

Ecthyma gangrenosum—rapidly progressive, necrotic cutaneous lesion **B** caused by *Pseudomonas* bacteremia. Typically seen in immunocompromised patients.

Treatments:

- Antipseudomonal penicillins in combination with β -lactamase inhibitor (eg, piperacillin-tazobactam)
- 3rd- and 4th-generation cephalosporins (eg, ceftazidime, cefepime)
- Monobactams
- Fluoroquinolones
- Carbapenems

Despite antipseudomonal activity, aminoglycoside monotherapy is avoided due to poor performance in acidic environments.

Salmonella vs Shigella

Both *Salmonella* and *Shigella* are gram $\ominus$ rods, non-lactose fermenters, oxidase $\ominus$, and can invade the GI tract via M cells of Peyer patches.

	<i>Salmonella typhi</i> (ty-Vi)	<i>Salmonella</i> spp. except <i>S typhi</i>	<i>Shigella</i>
RESERVOIRS	Humans only	Humans and animals	Humans only
SPREAD	Hematogenous spread	Hematogenous spread is rare	Cell to cell; no hematogenous spread
H ₂ S PRODUCTION	Yes	Yes	No
FLAGELLA	Yes (salmon swim)	Yes (salmon swim)	No
VIRULENCE FACTORS	Endotoxin; Vi capsule (pronounce “ty Vi ”)	Endotoxin	Endotoxin; Shiga toxin (enterotoxin)
INFECTIOUS DOSE (ID ₅₀)	High—large inoculum required; acid-labile (inactivated by gastric acids)	High	Low—very small inoculum required; acid stable (resistant to gastric acids)
EFFECT OF ANTIBIOTICS ON FECAL EXCRETION	Prolongs duration	Prolongs duration	Shortens duration (shortens <i>Shigella</i>)
IMMUNE RESPONSE	Primarily monocytes	PMNs in disseminated disease	Primarily PMN infiltration
GI MANIFESTATIONS	Constipation, followed by diarrhea	Diarrhea (possibly bloody)	Crampy abdominal pain → tenesmus, bloody mucoid stools (bacillary dysentery)
VACCINE	Oral vaccine contains live attenuated <i>S typhi</i> IM vaccine contains Vi capsular polysaccharide	No vaccine	No vaccine
UNIQUE PROPERTIES	Causes typhoid fever (salmon-colored truncal macular rash, abdominal pain, fever [pulse-temperature dissociation]; later GI ulceration and hemorrhage); treat with ceftriaxone or fluoroquinolone Carrier state with gallbladder colonization	Poultry, eggs, pets, and turtles are common sources Treatment is supportive; antibiotics are not indicated in immunocompetent individuals	4 F’s : f ingers, f lies, f ood, f eces In order of decreasing severity (less toxin produced): <i>S dysenteriae</i> , <i>S flexneri</i> , <i>S boydii</i> , <i>S sonnei</i> Invasion of M cells is key to pathogenicity; infectious dose is low

Yersinia enterocolitica

Gram $\ominus$ pleomorphic rod/coccobacillus with bipolar staining. Usually transmitted from pet feces (eg, cats, dogs), contaminated milk, or pork. Can cause acute bloody diarrhea, pseudoappendicitis (right lower abdominal pain due to mesenteric adenitis and/or terminal ileitis), reactive arthritis in adults.

Lactose-fermenting enteric bacteria

Fermentation of **lactose** → pink colonies on Mac**Con**key agar. Examples include *Citrobacter*, *E coli*, *Enterobacter*, *Klebsiella*, *Serratia*.

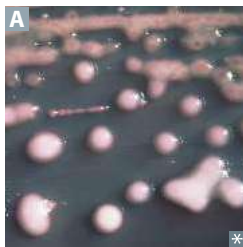
Mc**Cow**key **CEEKS** milk.

EMB agar—lactose fermenters grow as purple/black colonies. *E coli* grows colonies with a green sheen.

Escherichia coli

Gram $\ominus$, indole $\oplus$ rod. *E. coli* virulence factors: fimbriae (ie, P **p**ili)—cystitis and **p**yelonephritis; K capsule—pneumonia, neonatal meningitis; LPS endotoxin—septic shock.

STRAIN	TOXIN AND MECHANISM	PRESENTATION
Enteroinvasive <i>E. coli</i>	Microbe invades intestinal mucosa and causes necrosis and inflammation.	EIEC is I nvasive; dysentery. Clinical manifestations similar to <i>Shigella</i> .
Enterotoxigenic <i>E. coli</i>	Produces heat-labile and heat-stable enterotoxins. No inflammation or invasion.	ETEC ; T raveler's diarrhea (watery).
Enteropathogenic <i>E. coli</i>	No toxin produced. Adheres to apical surface, flattens villi, prevents absorption.	Diarrhea, usually in children (think EPEC and P ediatrics).
Enterohemorrhagic <i>E. coli</i>	O157:H7 is most common serotype in US. Often transmitted via undercooked meat, raw leafy vegetables. Shiga toxin causes hemolytic-uremic syndrome—triad of anemia, thrombocytopenia, and acute kidney injury due to microthrombi forming on damaged endothelium → mechanical hemolysis (with schistocytes on peripheral blood smear), platelet consumption → thrombocytopenia, and ↓ renal blood flow.	Dysentery (toxin alone causes necrosis and inflammation). Does not ferment sorbitol (vs other <i>E. coli</i>). EHEC associated with hemorrhage, hamburgers, hemolytic-uremic syndrome.

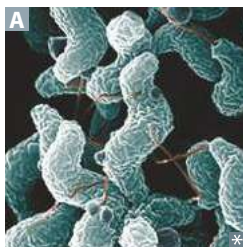
Klebsiella

Gram $\ominus$ rod; intestinal microbiota that causes lobar pneumonia; more common in patients with heavy alcohol use or with impaired host defenses. Very mucoid colonies **A** caused by abundant polysaccharide capsules. Dark red “currant jelly” sputum (blood/mucus).

Also cause of healthcare-associated UTIs. Associated with evolution of multidrug resistance (MDR).

ABCDE's of *Klebsiella*:

Aspiration pneumonia
aBscess in lungs and liver
“Currant jelly” sputum
Diabetes mellitus
EtOH overuse

Campylobacter jejuni

Gram $\ominus$, comma or S shaped (with polar flagella) **A**, oxidase $\oplus$, grows at **42°C** (“*Campylobacter* likes the **h**ot **c**ampfire”).

Major cause of bloody diarrhea, especially in children. Fecal-oral transmission through person-to-person contact or via ingestion of undercooked contaminated poultry or meat, unpasteurized milk. Contact with infected animals (dogs, cats, pigs) is also a risk factor.

Common antecedent to Guillain-Barré syndrome and reactive arthritis.

Proteus mirabilis

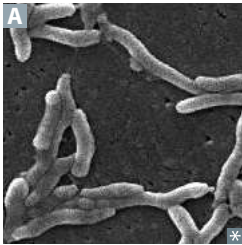
Gram $\ominus$, urease $\oplus$, facultative anaerobe, long flagellae with “swarming” motility.

Common cause of UTIs. Urease (virulence factor) hydrolyzes urea to carbon dioxide and ammonia → net increase in pH → promotes formation of struvite stones. Significant blockage of renal calyces results in branched stones called staghorn calculi.

Vibrio cholerae

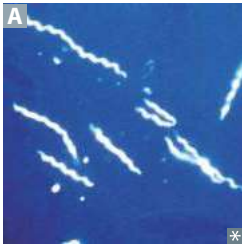
Gram $\ominus$, flagellated, comma shaped **A**, oxidase $\oplus$, grows in alkaline media. Endemic to developing countries. Produces profuse rice-water diarrhea via enterotoxin that permanently activates G_s , $\uparrow$ cAMP. Sensitive to stomach acid (acid labile); requires large inoculum (high ID_{50}) unless host has $\downarrow$ gastric acidity. Transmitted via ingestion of contaminated water or uncooked food (eg, raw shellfish). Treat promptly with oral rehydration solution.

Vibrio vulnificus—gram $\ominus$ bacillus, usually found in marine environments. Causes severe wound infections or septicemia due to exposure to contaminated sea water. Presents as cellulitis that can progress to necrotizing fasciitis in high-risk patients, especially those with high serum iron (eg, cirrhosis, hemochromatosis). Serious wound infection requires surgical debridement.

Helicobacter pylori

Curved, flagellated (motile), gram $\ominus$ rod **A** that is **triple** $\oplus$: catalase $\oplus$, oxidase $\oplus$, and urease $\oplus$ (can use urea breath test or fecal antigen test for diagnosis). Urease produces ammonia, creating an alkaline environment, which helps *H. pylori* survive in acidic mucosa. Colonizes mainly antrum of stomach; causes gastritis and peptic ulcers (especially duodenal). Risk factor for peptic ulcer disease, gastric adenocarcinoma, and MALT lymphoma.

Most common initial treatment is **triple** therapy: amoxicillin (metronidazole if penicillin allergy) + clarithromycin + proton pump inhibitor; antibiotics cure *P. pylori*. Bismuth-based quadruple therapy if concerned about macrolide resistance.

Spirochetes

Spiral-shaped bacteria **A** with axial filaments. Includes *Leptospira*, *Treponema*, and *Borrelia*. Only *Borrelia* can be visualized using aniline dyes (Wright or Giemsa stain) in light microscopy due to size. *Treponema* is visualized by dark-field microscopy or direct fluorescent antibody (DFA) microscopy.

Little Twirling Bacteria.

Jarisch-Herxheimer reaction—flu-like symptoms (fever, chills, headache, myalgia) after antibiotics are started due to host response to sudden release of bacterial antigens. Usually occurs during treatment of spirochetal infections.

Lyme disease

Caused by *Borrelia burgdorferi*, which is transmitted by the *Ixodes* deer tick **A** (also vector for *Anaplasma* spp. and protozoa *Babesia*). Natural reservoir is the mouse; deer are essential to tick life cycle but do not harbor *Borrelia*.

Common in northeastern United States.

Stage 1—early localized: erythema migrans (typical “bull’s-eye” configuration **B** is pathognomonic but not always present), flu-like symptoms.

Stage 2—early disseminated: secondary lesions, carditis, AV block, facial nerve (Bell) palsy, migratory myalgias/transient arthritis.

Stage 3—late disseminated: encephalopathy, chronic arthritis, peripheral neuropathy.

A Key **Lyme** pie to the **FACE**:

Facial nerve palsy (typically bilateral)

Arthritis

Cardiac block

Erythema migrans

Treatment: doxycycline (1st line); amoxicillin (pregnant patients, children < 8 years old); ceftriaxone if IV therapy required

Leptospira interrogans

Spirochete with hook-shaped ends found in water contaminated with animal urine.

Leptospirosis—flu-like symptoms, myalgias (classically of calves), jaundice, photophobia with conjunctival suffusion (erythema without exudate). Prevalent among surfers and in the tropics (eg, Hawaii).

Weil disease (icterohemorrhagic leptospirosis)—severe form with jaundice and azotemia from liver and kidney dysfunction, fever, hemorrhage, and anemia.

Syphilis

Caused by spirochete *Treponema pallidum*. Treatment: penicillin G.

Primary syphilis

Localized disease presenting with **painless** chancre. Use fluorescent or dark-field microscopy to visualize treponemes in fluid from chancre **A**. VDRL ⊕ in ~ 80% of patients.

Secondary syphilis

Disseminated disease with constitutional symptoms, maculopapular rash **B** (including palms **C** and soles), condylomata lata **D** (smooth, painless, wartlike white lesions on genitals), lymphadenopathy, patchy hair loss; also confirmable with dark-field microscopy. Serologic testing: VDRL/RPR (nonspecific), confirm diagnosis with specific test (eg, FTA-ABS). Secondary syphilis = systemic. Latent syphilis (⊕ serology without symptoms) may follow.

Tertiary syphilis

Gummas **E** (chronic granulomas), aortitis (vasa vasorum destruction), neurosyphilis (tabes dorsalis, “general paresis”), Argyll Robertson pupil (constricts with accommodation but is not reactive to light).

Signs: broad-based ataxia, ⊕ Romberg, Charcot joint, stroke without hypertension.

Congenital syphilis

Presents with facial abnormalities such as rhagades (linear scars at angle of mouth, black arrow in **F**), snuffles (nasal discharge, red arrow in **F**), saddle nose, notched (Hutchinson) teeth **G**, mulberry molars, and short maxilla; saber shins; CN VIII deafness.

To prevent, treat patient early in pregnancy, as placental transmission typically occurs after first trimester.

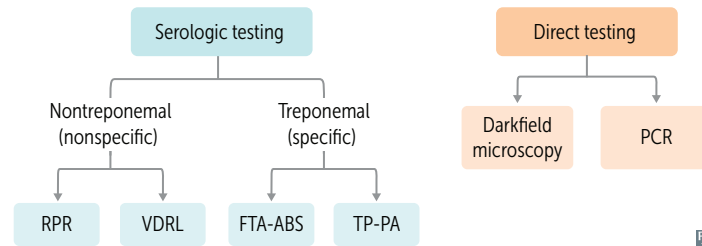


Diagnosing syphilis

VDRL and RPR detects nonspecific antibody that reacts with beef cardiolipin. Quantitative, inexpensive, and widely available test for syphilis (sensitive but not specific). Nontreponemal tests (VDRL, RPR) and direct testing revert to negative after treatment. Antibodies detected by treponemal tests (FTA-ABS, TP-PA) will remain positive.

False-Positive results on **VDRL** with:

- P**regnancy
- V**iral infection (eg, EBV, hepatitis)
- D**rugs (eg, chlorpromazine, procainamide)
- R**heumatic fever (rare)
- L**upus (anticardiolipin antibody) and **L**eprosy

**Chlamydiae**

Chlamydiae cannot make their own ATP. They are obligate intracellular organisms that cause mucosal infections. 2 forms:

- **E**lementary body (small, dense) is “infectious” and enters cell via endocytosis; transforms into reticulate body.
- **R**eticulate body replicates in cell by fission; reorganizes into elementary bodies.

Chlamydia trachomatis causes neonatal and follicular adult conjunctivitis **A**, nongonococcal urethritis, PID, and reactive arthritis.

Chlamydophila pneumoniae and *Chlamydophila psittaci* cause atypical pneumonia; transmitted by aerosol.

Chlamydial cell wall lacks classic peptidoglycan (due to reduced muramic acid), rendering β -lactam antibiotics ineffective.

Chlamys = cloak (intracellular).

C psittaci—has an avian reservoir (**p**arrots), causes atypical **p**neumonia.

Lab diagnosis: PCR, NAAT. Cytoplasmic inclusions (reticulate bodies) seen on Giemsa or fluorescent antibody–stained smear.

Treatment: doxycycline, azithromycin (for pregnant patients). Add ceftriaxone if concurrent gonorrhea testing is positive.

Chlamydia trachomatis* serotypes*Types A, B, and C**

Chronic infection, cause blindness due to follicular conjunctivitis in resource-limited areas.

ABC = **A**frica, **B**lindness, **C**hronic infection.

Types D–K

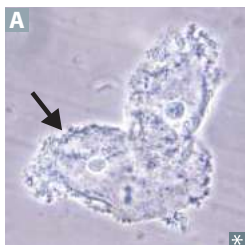
Urethritis/PID, ectopic pregnancy, neonatal pneumonia (staccato cough) with eosinophilia, neonatal conjunctivitis (1–2 weeks after birth).

D–K = everything else.

Neonatal disease can be acquired during vaginal birth if pregnant patient is infected.

Types L1, L2, and L3

Lymphogranuloma venereum—small, painless ulcers on genitals → swollen, painful inguinal lymph nodes that ulcerate (buboes). Treat with doxycycline.

Gardnerella vaginalis

A pleomorphic, gram-variable rod involved in bacterial vaginosis. Presents as a gray vaginal discharge with a fishy smell; nonpainful (vs vaginitis). Associated with sexual activity, but not sexually transmitted. Bacterial vaginosis is also characterized by overgrowth of certain anaerobic bacteria in vagina (due to ↓ lactobacilli). Clue cells (vaginal epithelial cells covered with *Gardnerella*) have stippled appearance along outer margin (arrow in **A**).

Amine whiff test—mixing discharge with 10% KOH enhances fishy odor.
Vaginal pH >4.5 during infection.
Treatment: metronidazole or clindamycin.

Zoonotic bacteria

Zoonosis—infectious disease transmitted between animals and humans.

SPECIES	DISEASE	TRANSMISSION AND SOURCE
<i>Anaplasma</i> spp	Anaplasmosis	<i>Ixodes</i> ticks (live on deer and mice)
<i>Bartonella</i> spp	Cat scratch disease, bacillary angiomatosis	Cat scratch
<i>Borrelia burgdorferi</i>	Lyme disease	<i>Ixodes</i> ticks (live on deer and mice)
<i>Borrelia recurrentis</i>	Relapsing fever	Louse (recurrent due to variable surface antigens)
<i>Brucella</i> spp	Brucellosis/undulant fever	Unpasteurized dairy; inhalation of or contact with infected animal tissue or fluids
<i>Campylobacter</i>	Bloody diarrhea	Feces from infected pets/animals; contaminated meats/foods/hands
<i>Chlamydophila psittaci</i>	Psittacosis	Parrots, other birds
<i>Coxiella burnetii</i>	Q fever	Aerosols of cattle/sheep amniotic fluid
<i>Ehrlichia chaffeensis</i>	Ehrlichiosis	<i>Amblyomma</i> (Lone Star tick)
<i>Francisella tularensis</i>	Tularemia	Ticks, rabbits, deer flies
<i>Leptospira</i> spp	Leptospirosis	Animal urine in water; recreational water use
<i>Mycobacterium leprae</i>	Leprosy	Humans with lepromatous leprosy; armadillo (rare)
<i>Pasteurella multocida</i>	Cellulitis, osteomyelitis	Animal bite, cats, dogs
<i>Rickettsia prowazekii</i>	Epidemic typhus	Human to human via human body louse
<i>Rickettsia rickettsii</i>	Rocky Mountain spotted fever	<i>Dermacentor</i> (dog tick)
<i>Rickettsia typhi</i>	Endemic typhus	Fleas
<i>Salmonella</i> spp (except <i>S typhi</i>)	Diarrhea (which may be bloody), vomiting, fever, abdominal cramps	Reptiles and poultry
<i>Yersinia pestis</i>	Plague	Fleas (rats and prairie dogs are reservoirs)

Rickettsial diseases and vector-borne illnesses

Treatment: doxycycline.

RASH COMMON

Rocky Mountain spotted fever

Rickettsia rickettsii, vector is tick. Despite its name, disease occurs primarily in the South Atlantic states, especially North Carolina. Rash typically starts at wrists **A** and ankles and then spreads to trunk, palms, and soles.

Classic triad—headache, fever, rash (vasculitis). **Palms** and **soles** rash is seen in **C**oxsackievirus **A** infection (hand, foot, and mouth disease), **R**ocky Mountain spotted fever, and 2° **S**yphilis (you drive **CARS** using your **palms** and **soles**).

Typhus

Endemic (fleas)—*R typhi*.
Epidemic (human body louse)—*R prowazekii*.
Rash starts centrally and spreads out, sparing palms and soles.

Rickettsii on the **wrists**, **typhus** on the **trunk**.

RASH RARE

Ehrlichiosis

Ehrlichia, vector is tick. **M**onocytes with morulae **B** (mulberrylike inclusions) in cytoplasm.

MEGA:

Monocytes = **E**hrlichiosis

Granulocytes = **A**naplasmosis

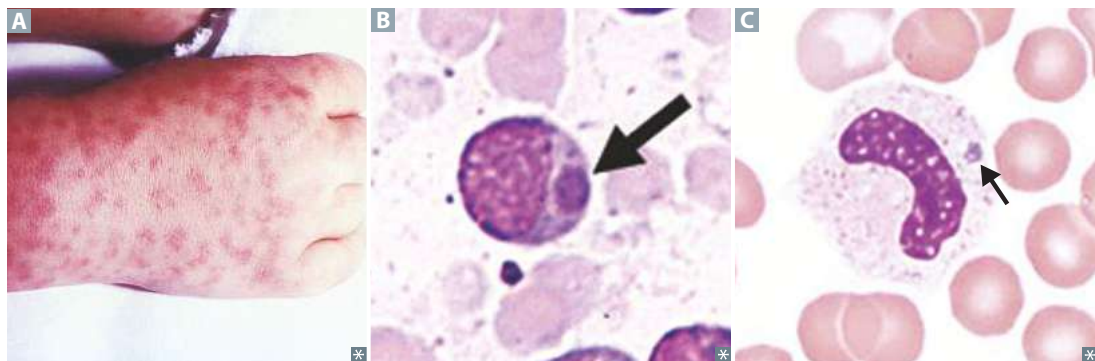
Anaplasmosis

Anaplasma, vector is tick. **G**ranulocytes with morulae **C** in cytoplasm.

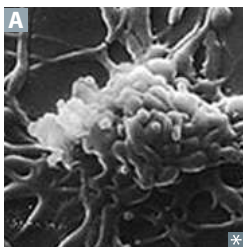
Q fever

Coxiella burnetii, no arthropod vector. Bacterium inhaled as aerosols from cattle/sheep amniotic fluid. Presents with headache, cough, flulike symptoms, pneumonia, possibly in combination with hepatitis. Common cause of culture ⊖ endocarditis.

Q fever is caused by a **Q**uite **C**omplicated **bug** because it has no rash or vector and its causative organism can survive outside in its endospore form. Not in the *Rickettsia* genus, but closely related.



Mycoplasma pneumoniae



Classic cause of atypical “walking pneumonia” (insidious onset, headache, nonproductive cough, patchy or diffuse interstitial infiltrate, macular rash).

Occurs frequently in those <30 years old; outbreaks in military recruits, prisons, colleges. Treatment: macrolides, doxycycline, or fluoroquinolone (penicillin ineffective since *Mycoplasma* has no cell wall).

Not seen on Gram stain. Pleomorphic **A**.

Bacterial membrane contains sterols for stability. Grown on Eaton agar.

CXR appears more severe than patient presentation. High titer of **cold** agglutinins (IgM), which can agglutinate RBCs. *Mycoplasma* gets **cold** without a **coat** (no cell wall).